Video-RAG: Visually-aligned Retrieval-Augmented Long Video Comprehension

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Abstract

Long video understanding remains a significant challenge for Large Vision-Language Models (LVLMs) due to context window limitations and the inefficiency of dense visual frame sampling. In this work, we propose *Video-RAG*, a training-free and plug-and-play pipeline that enhances long video comprehension by retrieving and injecting visually-aligned auxiliary texts—including optical character recognition (OCR), automatic speech recognition (ASR), and object detection (DET) information—into the input of any LVLM. Video-RAG operates in two stages: first, the LVLM decouples the user query into structured retrieval requests; second, a retrieval-augmented system constructs multimodal databases from the video and retrieves only the query-relevant textual information using a dense retriever with similarity-based filtering. Extensive experiments on three long video understanding benchmarks demonstrate the effectiveness of our approach. On Video-MME [3], Video-RAG yields an average improvement of 2.8% across seven LVLMs using only ∼2.0K auxiliary tokens per sample. Notably, applying Video-RAG to LLaVA-Video-72B [26] achieves 77.4%, surpassing GPT-4o [14] (77.2%). On MLVU [27] and LongVideoBench [21], Video-RAG establishes new state-of-the-art results among open-source models.

1 Introduction

Understanding long-form video content is a fundamental yet challenging task in multimodal AI. Videos spanning minutes to hours contain rich information distributed across visual, auditory, and textual modalities, making comprehensive understanding inherently difficult. Recent Large Vision-Language Models (LVLMs) such as GPT-4o [14] and Gemini-1.5-Pro [17] have demonstrated impressive capabilities in short video understanding, yet they still struggle with long videos due to two core limitations.

First, the context window of LVLMs imposes a hard constraint on the number of visual frames that can be processed simultaneously. As video duration increases, models must either subsample frames—risking the loss of critical information—or compress visual features, which can degrade fine-grained understanding. Second, increasing the sampling density leads to a large number of redundant visual tokens that consume context capacity without proportional information gain, as many frames in a long video contain repetitive or irrelevant content.

Existing approaches to address these challenges fall into two categories. The first line of work extends the context window through long-context fine-tuning [24, 23], enabling models to ingest more frames at the cost of expensive training. The second employs GPT-based agent frameworks that decompose videos into segments and query them iteratively, but these methods are computationally expensive and depend on proprietary APIs.

In this paper, we introduce *Video-RAG*, a fundamentally different approach that sidesteps both limitations. Our key insight is that much of the information needed to answer questions about long videos is not purely visual but rather multimodal in nature: spoken dialogue, on-screen text, and

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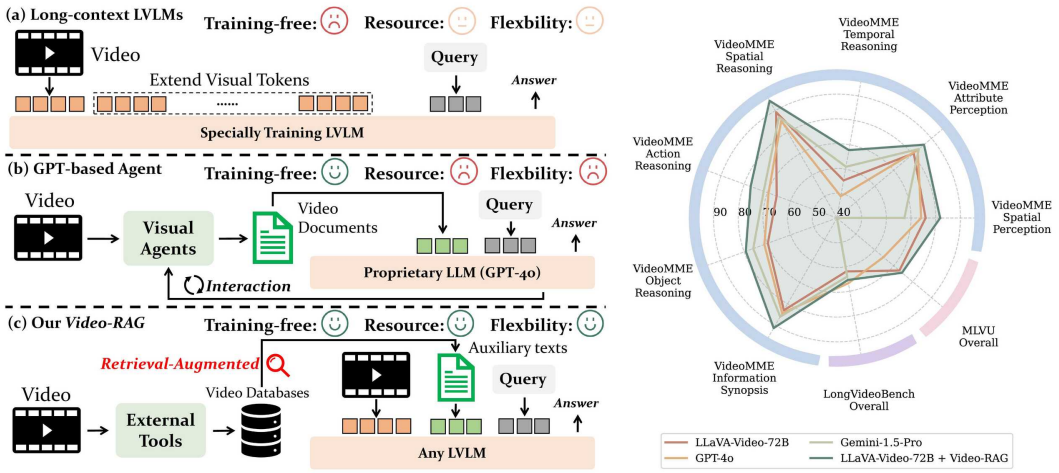

Figure 1: (Left) Two common approaches for long video understanding: extending the context window and agent-based systems. (Right) Our *Video-RAG* approach replaces redundant visual tokens with query-relevant auxiliary texts extracted from the video, achieving strong performance without additional training.

object identities can often be captured more compactly as text than as visual tokens. By replacing redundant visual frames with query-relevant textual information retrieved from the video, we can dramatically improve long video understanding without any additional training.

Video-RAG is a training-free, plug-and-play pipeline that can be applied to any LVLM. It operates in two stages. In the first stage, the LVLM itself decouples the user's query into a structured retrieval request specifying what audio, text, and object information to look for. In the second stage, Video-RAG constructs databases from the video's OCR, ASR, and object detection outputs, then retrieves only the query-relevant entries using a dense retriever [5] with efficient similarity search [7]. The retrieved auxiliary texts are prepended to the user's query, providing the LVLM with rich contextual information at a fraction of the token cost of processing additional visual frames.

Our contributions are as follows:

- We propose *Video-RAG*, a training-free, plug-and-play retrieval-augmented pipeline that enhances long video comprehension for any LVLM by injecting visually-aligned auxiliary texts.
- We design a query decoupling mechanism that leverages the LVLM itself to generate structured retrieval requests, enabling targeted extraction of OCR, ASR, and object detection information.
- Extensive experiments on Video-MME [3], MLVU [27], and LongVideoBench [21] demonstrate consistent improvements across seven open-source LVLMs, with LLaVA-Video-72B [26] reaching 77.4% on Video-MME to surpass GPT-4o [14].

2 Related Work

Long Video-Language Models. Recent efforts to extend LVLMs for long video understanding follow several directions. One approach increases the number of input frames through long-context fine-tuning: LongVA [24] and Long-LLaVA [23] extend the context window to handle hundreds of frames, while Video-XL [19] processes up to 256 frames. Another direction focuses on building stronger video-specialized LVLMs: LLaVA-Video [26] fine-tunes on large-scale video instruction data, VITA-1.5 [4] integrates audio understanding, and Video-CCAM [2] employs cross-modal attention mechanisms. Despite these advances, all approaches still face fundamental limitations in the number of visual tokens they can process, which motivates our text-augmentation strategy.

Retrieval-Augmented Generation for Vision. Retrieval-Augmented Generation (RAG) has been widely adopted in NLP to inject external knowledge into language models. Its application to vision tasks, however, remains relatively underexplored. In the video domain, prior work has primarily used retrieval to select relevant video clips or frames as input to models. Our approach differs fundamentally: rather than retrieving visual content, we retrieve textual information extracted from the video itself—OCR text, ASR transcripts, and object detection results—which provides complementary information that is difficult to capture through visual tokens alone.

Agent-Based Video Understanding. A complementary paradigm decomposes long video understanding into a multi-step inference pipeline. These agent systems typically partition a video into short segments, generate textual descriptions for each segment using a captioning model, and then retrieve or aggregate the resulting summaries at query time. While such systems can handle arbitrarily long videos, they introduce substantial latency due to multi-round model calls, often rely on expensive proprietary captioning APIs, and risk losing fine-grained visual detail during the summarization step. Video-RAG sidesteps these issues by extracting structured, modality-specific information (OCR, ASR, and object detection) as a one-time preprocessing step and performing a single lightweight retrieval pass at inference time, resulting in low latency and minimal additional cost per query.

Video Understanding with Auxiliary Modalities. Several works have explored using auxiliary modalities for video understanding. Whisper [16] enables high-quality speech transcription that can supplement visual analysis, while EasyOCR [6] extracts on-screen text. Object detection approaches such as APE [18] provide structured scene understanding. Video-ChatGPT [13] and ShareGPT4Video [1] leverage detailed video descriptions for training. However, these works typically use auxiliary information during training rather than at inference time. Video-RAG is unique in its training-free approach to combining OCR, ASR, and detection outputs through retrieval-augmented generation, selectively injecting only query-relevant information.

3 Method

We present *Video-RAG*, a training-free pipeline that augments any Large Vision-Language Model with retrieved auxiliary textual information extracted from videos. As illustrated in Figure 2, the pipeline consists of two main stages: query decoupling (Step 0) and database construction with retrieval (Step 1).

3.1 Step 0: Query Decoupling

Given a user query q about a video, we leverage the LVLM itself to decompose the query into a structured retrieval request. The model generates a JSON object with three fields:

- **ASR**: A text description of audio or subtitle content relevant to the query.
- **DET**: A list of up to five physical entities to detect in the video frames.
- **TYPE**: A list of information types needed from the scene, drawn from {"location", "number", "relation"}.

This query decoupling step requires no additional training—it relies on the LVLM's existing instruction-following capability to produce the structured output.

3.2 Step 1: Database Construction and Retrieval

Video-RAG constructs three complementary databases from the input video:

OCR Database. We apply EasyOCR [6] to extract on-screen text from all sampled video frames. Extracted text with confidence below 0.5 is discarded, and duplicate entries across consecutive frames are removed through deduplication. The resulting OCR entries are indexed by their frame timestamps.

ASR Database. We use Whisper-large [16] to transcribe the video's audio track in 30-second chunks. The transcribed segments are stored with their corresponding time intervals, capturing spoken dialogue, narration, and other audio content.

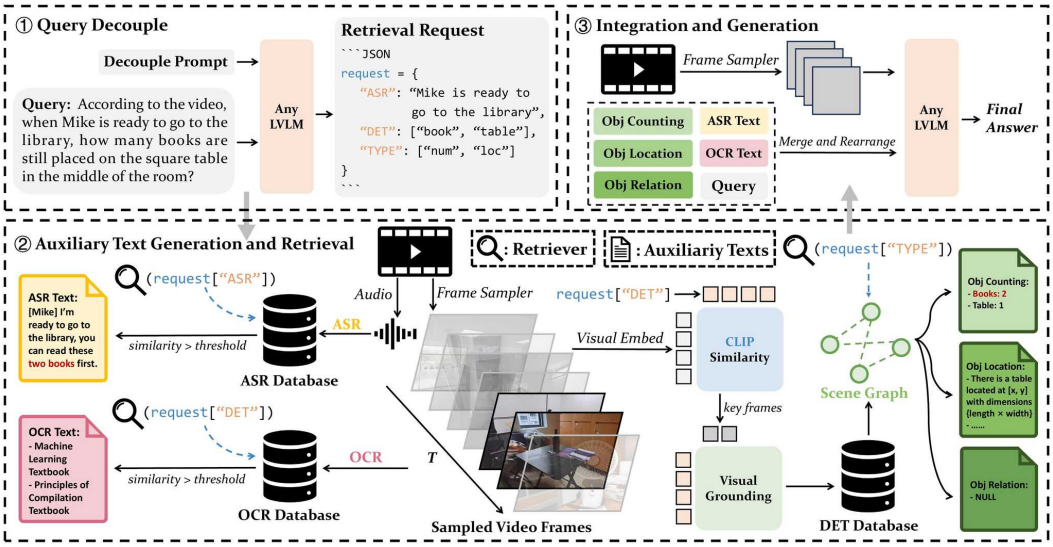

Figure 2: Overview of the *Video-RAG* pipeline. Given a user query, the LVLM first decouples it into structured retrieval requests (Step 0). Then, multimodal databases (OCR, ASR, DET) are constructed from the video and queried via dense retrieval to produce auxiliary texts that are prepended to the original query (Step 1).

Object Detection Database. For object detection, we employ a two-stage approach. First, we use CLIP-ViT-L/14-336 [15] to compute the similarity between each video frame and the entity descriptions from the DET field in the decoupled query. Frames with a CLIP similarity score above a threshold of 0.3 are selected as keyframes. Then, APE [18] is applied to these keyframes to detect objects and generate scene graphs that capture spatial relationships (left_of, right_of, above, below, overlaps), object locations, and counts.

Dense Retrieval. We use Contriever [5] to encode all database entries and the user query into dense vector representations. The vectors are L_2-normalized and indexed using FAISS [7] with an IndexFlatIP index for maximum inner-product (cosine similarity) search. The query vector is computed as the average of the embeddings of the original question and the specific retrieval request from the decoupled query. We perform range search to retrieve all documents with cosine similarity $\geq t$, where $t = 0.3$ serves as the retrieval threshold. This threshold-based approach, as opposed to top-k retrieval, naturally adapts the amount of retrieved information to the query's needs.

3.3 Auxiliary Text Formatting

The retrieved information is formatted and prepended to the user's original query in the order DET $\rightarrow$ ASR $\rightarrow$ OCR:

- **DET**: "Video has N frames in total, the detected objects' information in specific frames: Frame X: [scene graph]..."
- **ASR**: "Video Automatic Speech Recognition information (given in chronological order of the video): ..."
- **OCR**: "Video OCR information (given in chronological order of the video): ..."

The number of sampled frames is scaled by a factor $\beta = 3.0$, with a maximum of 32 frames. This formatted auxiliary text, averaging only $\sim$2.0K tokens per sample, provides the LVLM with rich contextual information while consuming significantly fewer tokens than full subtitle injection ($\sim$3.0K tokens).

4 Experiments

4.1 Experimental Setup

Benchmarks. We evaluate Video-RAG on three established long video understanding benchmarks: (1) **Video-MME** [3], which contains videos categorized by duration (short, medium, long) and evaluates models with and without subtitle information; (2) **MLVU** [27], a multiple-choice benchmark designed to assess understanding of long videos; and (3) **LongVideoBench** [21], which tests models on the validation set with diverse long video comprehension tasks.

Models. We apply Video-RAG to seven open-source LVLMs spanning different architectures and scales: Video-LLaVA [10] (7B), LLaVA-NeXT-Video [25] (7B), VITA-1.5 [4] (7B), LongVA [24] (7B), Long-LLaVA [23] (7B), Qwen2-VL [20] (72B), and LLaVA-Video [26] (72B). We compare against proprietary models GPT-4o [14] and Gemini-1.5-Pro [17]. All open-source results are our reproductions.

Implementation Details. For retrieval, we use Contriever [5] (facebook/contriever) as the dense encoder and FAISS [7] IndexFlatIP for similarity search. The retrieval threshold is set to $t = 0.3$, the CLIP similarity threshold for keyframe selection is 0.3, the frame scaling factor $\beta = 3.0$, and the maximum number of frames is 32. For ASR, we use Whisper-large [16] with 30-second audio chunks. OCR is performed using EasyOCR [6] with a confidence threshold of 0.5. Object detection uses APE [18] on CLIP-selected keyframes.

4.2 Main Results on Video-MME

Table 1 presents the comprehensive results on Video-MME. Video-RAG achieves consistent improvements across all seven open-source LVLMs, with an average gain of 2.8% over the subtitle-enhanced baselines. Notably, the improvements are most pronounced for models with smaller context windows: LongVA achieves +6.0% and Long-LLaVA achieves +4.8%, suggesting that Video-RAG is particularly effective at compensating for limited visual context.

Among the 7B models, Long-LLaVA with Video-RAG reaches 62.6%, a substantial improvement over its 57.8% baseline with subtitles. For larger models, Qwen2-VL-72B improves from 71.9% to 72.9%, and LLaVA-Video-72B improves from 75.9% to 77.4%. This latter result is particularly significant: **LLaVA-Video-72B with Video-RAG (77.4%) surpasses the proprietary GPT-4o (77.2%)**, demonstrating that our training-free approach can close the gap between open-source and proprietary models.

The improvements are consistent across video durations (short, medium, long), with larger gains typically observed on medium and long videos where auxiliary textual information provides greater complementary value to the limited visual frames.

4.3 Results on MLVU and LongVideoBench

Table 4 and Table 5 present results on MLVU and LongVideoBench, respectively. On MLVU, applying Video-RAG to LLaVA-Video-7B improves performance from 70.8% to 72.4% (+1.6%), and LLaVA-Video-72B reaches 73.8% (+0.7%), establishing a new state-of-the-art among all models including Oryx-1.5-32B [12] (72.3%) and surpassing GPT-4o [14] (64.6%) by a large margin.

On LongVideoBench, Video-RAG boosts LLaVA-Video-7B from 56.6% to 58.7% and LLaVA-Video-72B from 61.9% to 65.4%. The 72B result surpasses Gemini-1.5-Pro [17] (64.0%) and approaches GPT-4o (66.7%), further validating the generalizability of our approach across diverse benchmarks.

A finer-grained breakdown of LongVideoBench reveals that gains are largest on temporal reasoning and event localization subtasks, where ASR transcripts anchor query-relevant moments in the video timeline. On spatial understanding questions, the DET component contributes most, as structured scene graphs describing object positions and relationships provide information that is otherwise difficult to recover from subsampled visual frames. OCR proves particularly valuable for videos containing slides, captions, or on-screen notifications, where critical content appears as text. These complementary strengths across subtask categories confirm that Video-RAG's three retrieval modalities address distinct information needs rather than overlapping redundantly.

Table 1: Performance on the Video-MME [3] benchmark in without subtitles (w/o S), with subtitles (w/ S) and equipped with our Video-RAG (Ours), **Frames** and **Gain** means the input frame number and performance gain by applying Video-RAG compared to the baseline with subtitles. By applying Video-RAG to seven LVLMs, we observed an average performance improvement of 2.8% by adding only an average of ∼2.0K auxiliary texts compared to ∼3.0K full-subtitled tokens per sample. In particular, we perform better when applying Video-RAG with 72B LLaVA-Video [26] than the proprietary method GPT-4o [14] (77.4% vs. 77.2%). All results are our republication.

Model	Params	Frames	Short			Medium			Long			Overall			Gain
			w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	
Proprietary LVLMs															
GPT-4o [14]	-	384	80.0	82.8	-	70.3	76.6	-	65.3	72.1	-	71.9	77.2	-	-
Gemini-1.5-Pro [17]	-	0.5 fps	81.7	84.5	-	74.3	81.0	-	67.4	77.4	-	75.0	81.3	-	-
Open-Source LVLMs															
Video-LLaVA [10]	7B	8	45.3	46.1	49.5	38.0	40.7	43.0	36.2	38.1	42.5	39.9	41.6	45.0	+3.4
LLaVA-NeXT-Video [25]	7B	16	49.4	51.8	56.6	43.0	46.4	47.4	36.7	44.9	46.0	43.0	47.7	50.0	+2.3
VITA-1.5 [4]	7B	16	67.0	69.9	71.0	54.2	55.7	55.4	47.1	50.4	52.4	56.1	58.7	59.6	+0.9
LongVA [24]	7B	128	61.1	61.2	66.1	50.4	53.8	60.4	46.2	52.9	59.4	52.6	56.0	62.0	+6.0
Long-LLaVA [23]	7B	64	61.9	62.4	67.1	51.4	56.2	60.4	45.4	54.7	60.1	52.9	57.8	62.6	+4.8
Qwen2-VL [20]	72B	32	75.0	76.7	77.4	63.3	69.9	70.2	56.3	69.2	71.0	64.9	71.9	72.9	+1.0
LLaVA-Video [26]	72B	64	80.7	81.8	82.8	68.7	73.8	76.3	62.1	72.2	73.1	70.3	75.9	77.4	+1.5

Table 2: The overall performance in the multiple-choice task of the MLVU [27] benchmark. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
GPT-4o [14]	-	0.5 fps	64.6
Open-Source LVLMs			
VITA-1.5 [4]	7B	16	60.4
Video-CCAM [2]	14B	96	63.1
Video-XL [19]	7B	256	64.9
Aria [8]	25.3B	256	70.6
LLaVA-Video* [26]	7B	64	70.8
Oryx-1.5 [12]	32B	128	72.3
LLaVA-Video* [26]	72B	64	73.1
LLaVA-Video + Video-RAG	7B	64	72.4
LLaVA-Video + Video-RAG	72B	64	**73.8**

Table 3: The overall performance on the validation set of LongVideoBench [21]. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
Gemini-1.5-Pro [17]	-	256	64.0
GPT-4o [14]	-	256	**66.7**
Open-Source LVLMs			
VideoChat2-Mistral [9]	7B	8	39.3
ShareGPT4Video [1]	7B	8	39.7
LLaVA-Next-Mistral [11]	7B	8	49.1
PLLaVA [22]	34B	16	53.2
VITA-1.5 [4]	7B	16	53.6
LLaVA-Video* [26]	7B	64	56.6
LLaVA-Video* [26]	72B	64	61.9
LLaVA-Video + Video-RAG	7B	64	58.7
LLaVA-Video + Video-RAG	72B	64	65.4

5 Analysis

5.1 Ablation on Auxiliary Text Components

To understand the contribution of each component, we conduct an ablation study using Long-LLaVA-7B [23] on Video-MME, shown in the left panel of Table 4. Starting from a 52.0% baseline without any auxiliary text, we observe several findings.

First, ASR provides the largest individual gain. Without retrieval, adding ASR alone improves performance to 57.4% (+5.4%), while OCR alone contributes less. Combining all three components (DET+OCR+ASR) without retrieval yields 59.8%.

Second, retrieval-augmented generation further improves performance. With retrieval enabled, the full DET+OCR+ASR configuration achieves 62.1%, a +2.3% improvement over the non-retrieval version (59.8%). This demonstrates that selectively retrieving query-relevant information is more effective than injecting all extracted information indiscriminately.

Third, each component contributes complementary information. Comparing retrieval-based configurations, DET+ASR (61.5%) and OCR+ASR (62.0%) both perform well, but the full combination (62.1%) achieves the best overall result.

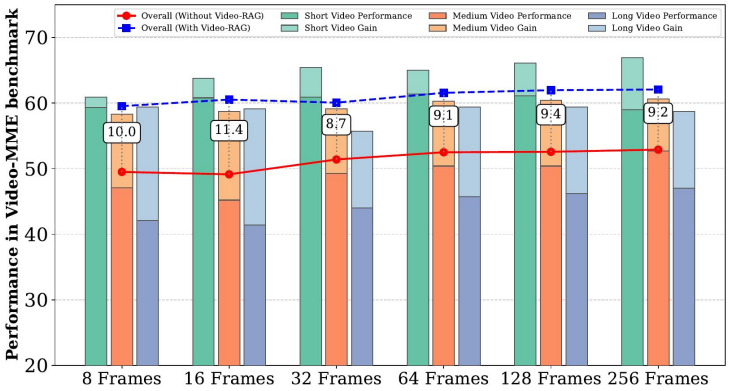

Figure 3: Performance gain of Video-RAG with different sampling frame rates on Video-MME. Video-RAG provides the largest improvements at lower frame rates, where visual information is sparse.

5.2 Effect of Retrieval Threshold

The right panel of Table 5 analyzes the impact of the retrieval threshold t on performance, token count, and inference time. Setting $t = 0.3$ achieves the best overall accuracy (62.1%) with only 1.9K tokens and 11 seconds per question. This represents a favorable trade-off: compared to $t = 0.0$ (no filtering), which uses 3.6K tokens and 36 seconds to achieve 62.0%, our threshold reduces token count by 47% and inference time by 69% while slightly improving accuracy. Higher thresholds ($t \geq 0.4$) aggressively filter information, causing performance degradation—$t = 0.5$ drops to 56.1% and $t = 1.0$ reverts to the no-augmentation baseline of 52.0%. The "rnd" row shows that random retrieval with the same token budget (1.9K) only achieves 59.1%, confirming the importance of similarity-based retrieval over random selection.

Table 4: Results on combinations of different auxiliary texts in Video-MME [3] when using Long-LLaVA-7B [23] as the LVLM.

RAG	DET	OCR	ASR	Short	Medium	Long	Overall
				60.3	51.4	44.1	52.0
			✓	62.2	55.4	54.4	57.4
		✓	✓	64.0	56.2	55.0	58.4
	✓		✓	63.0	57.3	**56.4**	58.9
	✓	✓	✓	**64.3**	**58.8**	56.3	**59.8**
✓	✓			61.4	51.9	45.2	52.9
✓		✓		63.2	53.2	46.3	54.3
✓			✓	65.1	59.1	**60.7**	61.6
✓	✓	✓		64.1	54.6	48.4	55.7
✓	✓		✓	64.9	59.0	**60.7**	61.5
✓		✓	✓	66.3	**60.3**	59.3	62.0
✓	✓	✓	✓	**66.4**	60.2	59.8	**62.1**

Table 5: Performance with different thresholds of retrieval on Video-MME [3] when using Long-LLaVA-7B [23] as the LVLM. **#Token** and **Time** denote the total token number of the auxiliary texts and the average inference time per question, respectively.

t	#Token	Time	Short	Medium	Long	Overall
0.0	3.6K	36s	**67.6**	59.4	59.1	62.0
0.1	3.4K	30s	67.0	59.7	59.1	61.9
0.2	2.7K	18s	66.0	**60.2**	59.2	61.8
0.3	1.9K	11s	66.4	**60.2**	**59.8**	**62.1**
0.4	0.8K	8s	65.6	58.0	58.3	60.6
0.5	0.3K	7s	63.1	54.9	50.2	56.1
1.0	0.0K	6s	60.3	51.4	44.1	52.0
rnd	1.9K	11s	65.7	55.8	56.0	59.1

5.3 Impact of Frame Sampling Rate

Figure 3 shows the performance gain of Video-RAG at different frame sampling rates on Video-MME. The improvement is most substantial at lower frame rates, where the visual information is sparse and auxiliary texts provide critical complementary signals. As the frame rate increases, the marginal gain from Video-RAG decreases but remains positive, indicating that textual information provides value beyond what additional visual frames capture.

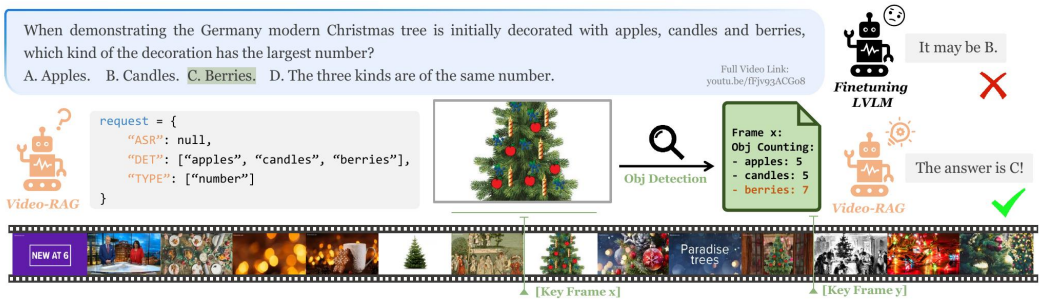

Figure 4: Qualitative result on Video-MME with LLaVA-Video [26]. Video-RAG provides auxiliary textual information that enables the model to correctly answer the question.

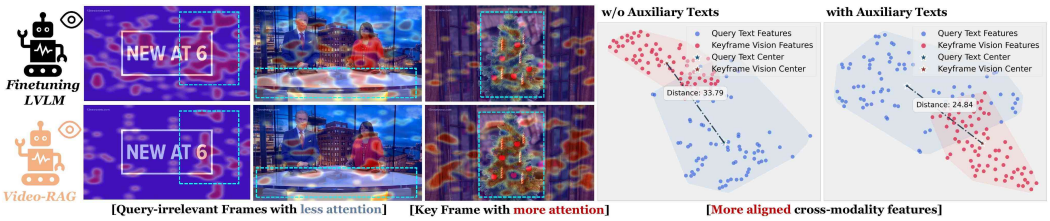

Figure 5: Grad-CAM attention maps and t-SNE visualizations of query and keyframe features. Video-RAG improves the alignment between query representations and relevant visual content.

5.4 Qualitative Analysis

Figure 4 presents a qualitative example from Video-MME using LLaVA-Video [26]. Without Video-RAG, the model fails to answer the question correctly due to insufficient visual context. With Video-RAG, the retrieved auxiliary texts provide the critical information—such as spoken dialogue captured by ASR or on-screen text captured by OCR—that enables the model to arrive at the correct answer. This illustrates how Video-RAG compensates for the limitations of visual-only processing by supplying complementary textual evidence.

5.5 Feature Visualization

To gain deeper insight into how Video-RAG influences model behavior, we visualize the query and keyframe features using Grad-CAM and t-SNE in Figure 5. The Grad-CAM visualizations show that Video-RAG helps the model attend to more relevant regions of the video frames. The t-SNE plots demonstrate that the retrieved auxiliary texts help align the query representations with the relevant keyframe features in the embedding space, facilitating more accurate matching between questions and visual content.

6 Conclusion

We presented *Video-RAG*, a training-free and plug-and-play retrieval-augmented pipeline for long video comprehension. By extracting multimodal auxiliary information (OCR, ASR, and object detection) from videos and selectively retrieving query-relevant texts via dense retrieval, Video-RAG provides LVLMs with rich contextual signals using only ∼2.0K tokens on average. Experiments across three benchmarks—Video-MME, MLVU, and LongVideoBench—demonstrate consistent improvements across seven open-source LVLMs. Notably, LLaVA-Video-72B with Video-RAG achieves 77.4% on Video-MME, surpassing GPT-4o (77.2%), and establishes new state-of-the-art results on MLVU and LongVideoBench among open-source models. Our approach demonstrates that leveraging textual information inherent in videos through retrieval-augmented generation is a simple yet effective strategy for advancing long video understanding. Looking ahead, future work could explore learning adaptive retrieval thresholds conditioned on query complexity, integrating

additional modalities such as optical flow for action-heavy videos, or combining Video-RAG with efficient visual token compression to further reduce inference cost.

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